SPARC - SQ

Spectral Quadrature method

User guide

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Contributors
Citation
Acknowledgements

Contributors

- **Phanish Suryanarayana** (PI)
  - **Xin Jing**: Code infrastructure
  - **Abhiraj Sharma**: Initial development
  - **Phanisri P. Pratapa**: Initial development
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Citation

If you publish work using/regarding SPARC-SQ, please cite the following articles, in addition to SPARC citations:

- https://doi.org/10.1016/j.cpc.2015.11.005 (initial implementation),
  https://doi.org/10.1016/j.cplett.2013.08.035 (formulation),
  https://doi.org/10.1007/978-3-031-22340-2_12 (detailed mathematical formulation),

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Comments

The code will fail with the following options and the related input options are listed.

- Polarized calculation: `SPIN_TYP`.
- K-point calculation: `KPOINT_GRID`, `KPOINT_SHIFT`.
- Dirichlet boundary condition in any direction: `BC`
- Define number of states/orbitals: `NSTATES`
- Print eigenvalues into file: `PRINT_EIGEN`

Input file options

Spectral Quadrature

SQ_AMBIENT_FLAG | SQ_HIGHT_FLAG | SQ_RCUT | SQ_NPL_G | SQ_HIGHT_GAUSS_HYBRID_MEM | SQ_TOL_OCC | NP_DOMAIN_SQ_PARAL

SPDFT

SP_FLAG | SP_SMEARING | SP_TOL_OCC | PRINT_SPDFT | SP_ISESPLIT_CONST

Spectral Quadrature

SQ_AMBIENT_FLAG

Type

Integer

Unit

No unit

Default

0

Example

```
SQ_AMBIENT_FLAG: 1
```

Description

Flag to turn on SQ method

Remark

SQ method can not be turned on simultaneously with CS, SQ3, hybrid functionals.

SQ_HIGHT_FLAG

Type

Integer

Unit

No unit

Default

0

Example

```
SQ_HIGHT_FLAG: 1
```

Description

Flag to turn on SQ method for High Temperature

Remark

At High Temperature, spin-polarized calculation is not supported and high memory usage is always enabled. To use Low Temperature SQ method, please turn it off by setting SQ_HIGHT_FLAG: 0.

SQ_RCUT

Type

Double

Unit

Bohr

Default

None

Example

SQ_RCUT: 2.0

Description

Truncation or localization radius

Remark

SQ_RCUT must be specified if SQ is turned on.

SQ_NPL_G

Type

Integer

Unit

No unit

Default

None

Example

SQ_NPL_G: 24

Description

Degree of polynomial for Gauss Quadrature.

Remark

SQ_NPL_G must be specified if SQ is turned on.

SQ_HIGHT_GAUSS_HYBRID_MEM

Type

String

Unit

No unit

Default

LOW

Example

SQ_HIGHT_GAUSS_HYBRID_MEM: HIGH

Description

Flag for memory option when using Gauss quadrature at High Temperature for saving the exact exchange potential when using hybrid functionals.

SQ_TOL_OCC

Type

Double

Unit

No unit

Default

10^{-6}

Example

SQ_TOL_OCC: 1E-5

Description

Tolerance for occupation corresponding to maximum eigenvalue.

NP_DOMAIN_SQ_PARAL

Type

Integer

Unit

No unit

Default

Automatically optimized

Example

`NP_DOMAIN_SQ_PARAL`: 3 3 2

Description

Dimensions of the 3D Cartesian topology for SQ method.

Remark

This option is for development purpose. It's better to let SPARC choose the parallization parameters in practice.

Spectral partitioned DFT

SP_FLAG

Type

Integer

Unit

No unit

Default

0

Example

SP_FLAG: 1

Description

Flag to turn on SP method.

Remark

SP method cannot be turned on simultaneously with SQ, CS, SQ3, hybrid functionals.

SP_SMEARING

Type

Double

Unit

Ha

Default

0.01

Example

SP_SMEARING: 0.05

Description

Regularization parameter to ensure numerical stability of SPDFT method.

Remark

To avoid its effect on the physics of the problem, it is kept approx. 50x smaller than the Fermi-Dirac smearing.

SP_TOL_OCC

Type

Double

Unit

No unit

Default

1e-12

Example

SP_TOL_OCC: 1e-13

Description

Allows obtaining the highest-energy plane wave in the spdft method, such that its occupation is less than this value.

PRINT_SPDFT

Type

Integer

Unit

No unit

Default

0

Example

```
PRINT_SPDFT: 1
```

Description

flag for printing relevant spDFT data into .spDFT file.

Remark

Must be turned on to obtain splitting energies of a spdft calculation.

SP_ISESPLIT_CONST

Type

Integer

Unit

No unit

Default

0

Example

```
SP_ISESPLIT_CONST: 1
```

Description

Flag to specify if the splitting energies in spdft method need to be kept constant while performing self-consistency. The constant value is read from .spDFT file.